5.4

Asymptotic Properties of MLEs

MLE intution and large-sample properties

The quality of estimator should improve as our sample size increases.
MLEs have good large-sample properties.

Definition 5.4.1

An unbiased estimator who variance attains the CRLB is said to be **efficient**.
The efficiency of an estimator T of $\tau(\theta)$ is the ratio
$\dfrac{\mathrm{CRLB}_\theta}{\mathrm{var}(T)}$.
An efficient estimator has efficiency 1.

MLEs have nice properties

Under certain regularity conditions, the MLE of θ, $\hat{\theta}_n$, will have nice properties:

- $\hat{\theta}_n$ **exists** and is **unique**.
- $\hat{\theta}_n$ is a **consistent** estimator of θ, i.e., $\hat{\theta}_n \xrightarrow{P} \theta$.
- $\hat{\theta}_n$ is **asymptotically unbiased** for θ, i.e., $\lim_{n\to\infty} E[\hat{\theta}_n] = \theta$.
- $\hat{\theta}_n$ is **asymptotically efficient**, i.e., $\lim_{n\to\infty} \frac{\mathrm{CRLB}_\theta}{\mathrm{var}(\hat{\theta}_n)} = 1$.
- $\hat{\theta}_n$ is **asymptotically normal**, i.e., $\hat{\theta}_n \overset{asymp.}{\sim} N(\theta, \mathrm{CRLB}_\theta)$.

Regularity Conditions for MLEs

MLEs are consistent under the following assumptions (Casella and Berger, 2002):

- A1: **Correct Model**. $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} f(x;\theta)$, i.e., that is their true distribution.
- A2: **Identifiability**. $\theta_1 \neq \theta_2 \implies f(x;\theta_1) \neq f(x;\theta_2)$.
- A3: **Differentiability**. $f(x;\theta)$ is differentiable in θ.
- A4: **Interior parameter**. The true parameter value θ_0 is an interior point of the parameter space Θ.

Regularity Conditions for MLEs (cont)

To prove asymptotic normality and efficiency of the MLEs, we also need conditions A5 and A6:

- A5: **Smoothness (regularity)**.
  - The density $f(x;\theta)$ is three times differentiable with respect to θ and continuous, i.e., $\frac{\partial^3}{\partial\theta^3} f(x;\theta)$ exists and is continuous.
  - $\int f(x;\theta)\,dx$ can be differentiated three times under the integral sign, i.e., $\frac{\partial^3}{\partial\theta^3} \int f(x;\theta)\,dx = \int \frac{\partial^3}{\partial\theta^3} f(x;\theta)\,dx$.

Regularity Conditions for MLEs (cont)

- A6: **Boundedness**. For all $\theta \in \Theta$, there exists a $c > 0$ and a function $M(x)$ such that
  $\left| \frac{\partial^3 \log f(x;\theta)}{\partial \theta^3} \right| \leq M(x)$
  with $E_{\theta_0}[M(X)] < \infty$ for all $|\theta - \theta_0| < c$ and all x in the support of X.

5.5

Uniformly Minimum Variance Unbiased Estimators (UMVUEs)

Definition 5.5.1

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} f(x;\theta)$. The **uniformly minimum variance unbiased estimator (UMVUE)** for θ is the unbiased estimator that has the smallest variance for all unbiased estimators.
"Uniformly" in θ means that this is true for all values of θ.
We will learn how to find and built UMVUEs in certain contexts.

5.5.1

Sufficient Statistics

Definition 5.5.2

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} f(x;\theta)$. $S = s(\vec{X})$ is a **sufficient** statistic for estimating θ if the conditional distribution of $\vec{X}$ given S does not depend on θ.
Informally, if you know S, then seeing $\vec{X}$ does not give you any more information about θ.

Sufficient statistics in practice

See Example 5.5.1 for manually verifying a statistic is sufficient.
In practice, we find sufficient statistics using Neyman's Factorization Theorem

Neyman's Factorization Theorem

If
$f(\vec{x};\theta) = h(\vec{x})g(s(\vec{x});\theta),$
where h does not depend on θ and g only depends on $\vec{x}$ through $s(\vec{x})$,
then
$S = s(\vec{x})$
is sufficient for estimating θ.

Example 5.5.2

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Bernoulli}(p)$. Determine a sufficient statistic for p.

Example 5.5.2 (cont)

Example 5.5.2 (cont)

Comments

- Sufficient statistics are not unique.
- $\vec{x}$ is a sufficient statistic but doesn't compress the data at all.
- We can always add more information and it will still be sufficient.
- We want to "compress" the data as much as possible, which foreshadows the concept of minimal sufficient statistics.
- Being sufficient for θ implies being sufficient for $\tau(\theta)$.

Example 5.5.3

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \mathrm{Uniform}(0, \theta)$. Find a sufficient statistic for θ.

Example 5.5.3 (cont)

Example 5.5.3 (cont)

5.5.2

Conditional Expectation and Variance

Conditional expectation and variance notes

Let X and Y be random variables.

- $E[X]$ is a constant.
- $E[X \mid Y = y]$ is a function of y.
- $E[X \mid Y]$ is a random variable.
- $E[E[X \mid Y]] = E[X]$ (the law of iterated expectation).
- $\mathrm{var}[X] = \mathrm{var}[E[X \mid Y]] + E[\mathrm{var}[X \mid Y]]$.